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Pinyin-Guided Chinese Speech Recognition with Large Language Model

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Abstract

While LLM-based automatic speech recognition (LLM-ASR) has demonstrated efficacy through direct acoustic-to-text mapping, its implicit alignment often fails to capture phonetic relationships in Chinese, leading to pronunciation confusion and homophone errors. This paper proposes Pinyin-Guided ASR (PYG-ASR), which innovatively modifies the LLM-ASR to simultaneously map acoustic features to both Pinyin and text tokens, enhancing linguistic representation. PYG-ASR leverages the generated Pinyin alongside text for error correction, prompting text LLM to refine transcriptions without finetuning. Furthermore, error correction phase inherently enables context biasing by filtering bias phrases through Pinyin matching and incorporating them into the prompt. Experiments show that PYG-ASR reduces CER by 25% on the AISHELL-1 test set. Additionally, our approach shows a 49.2% CER reduction relatively for bias phrases on the AISHELL-1 test set after contextual bias.

Index Terms: large language model, speech recognition, error correction, contextual biasing

1. Introduction

Recently, large language models (LLMs) [1–4] have exhibited remarkable capabilities across a range of natural language processing tasks, often requiring little to no finetuning on the downstream tasks. Building on these advancements, there has been a growing effort to extend LLM capabilities to various speech tasks such as audio captioning, speech translation, automatic speech recognition (ASR) [5–8]. Among these applications, ASR has garnered significant attention. A common approach involves combining pretrained speech encoders (e.g., Whisper encoder [9], HuBERT [10]) with text LLMs to create LLM-ASR models capable of handling ASR tasks [5, 11–14].

In LLM-ASR, acoustic features are typically mapped directly onto text tokens, bypassing intermediate phoneme representations. However, this approach may introduce biases in languages with rich phonetic complexity, such as Chinese. In Chinese, for instance, a small number of characters originate as pictographs and ideographs, but the vast majority are what are called phono-semantic compounds, which involve an element of pronunciation in their meaning. [15]. This indicates that there is no direct correspondence between the pronunciation and the written form of Chinese characters. The presence of tonal variations and a high number of homophones makes it essential to explicitly model pronunciation features, a capability inherently absent in most LLM-ASR models. This lack of phonetic modeling poses one of the main challenges for Chinese ASR.

Chinese Pinyin, often referred to simply as Pinyin, which represents the syllables of Chinese characters and reflects their pronunciation information, serves as a crucial tool for bridging this gap. It is widely used in China for language teaching and as a common method for entering Chinese characters on computers and smartphones. Directly mapping acoustic features to Chinese characters without the aid of Pinyin is like trying to write Chinese characters without the guidance of Pinyin. While it's not impossible, it can be more challenging and prone to transcription errors. Learning Pinyin first can provide helpful phonetic guidance that makes the process of learning and writing Chinese characters more efficient and accurate.

Previous research has highlighted the advantages of incorporating Pinyin in ASR tasks [16], mainly focusing on error correction using Pinyin or pretraining LLMs on Pinyin sequences to generate corresponding Chinese characters [17]. However, there has been little exploration of how the LLM-ASR model can directly generate Pinyin and Chinese characters during ASR task training. In this paper, we propose PYG-ASR, a novel Chinese ASR framework incorporating Pinyin guidance from the very beginning, with the aim to address the phonetic complexity in Chinese. Our specific approaches are as follows:

i Inspired by CoT-ST [18] which decomposes speech translation into sequential steps of speech recognition and translation, enhancing translation accuracy, we decompose Chinese speech recognition into Pinyin and Chinese character recognition by simultaneously mapping acoustic features to both Pinyin and text tokens. Unlike traditional LLM-ASR models that directly map acoustic features to text tokens, our approach explicitly models the pronunciation characteristics of Chinese, enhancing linguistic representation.

ii We introduce a Pinyin-Guided Error Correction (PYGEC) approach that utilizes the one-best hypothesis from LLM-ASR generation, including Chinese characters and Pinyin, by adopting a chain-of-thought prompting with the text LLM, leading to a significant reduction in Character Error Rate (CER).

iii In addition, PYGEC can effectively enhance contextual biasing by filtering bias phrases through Pinyin matching and utilizing these filtered phrases to prompt the text LLM, resulting in improved handling of specific contexts and enhanced recognition accuracy.

We validated the effectiveness of our proposed method on the AISHELL-1 [19] Chinese corpus. Compared with the baselines, PYG-ASR achieves considerable improvement on both overall and bias phrase accuracy.

2. Proposed Method

2.1. Pinyin System

Pinyin is a romanization system for Mandarin Chinese, consisting of initials, finals, and tones. While various Pinyin systems may exhibit minor variations in the number of initials and finals,

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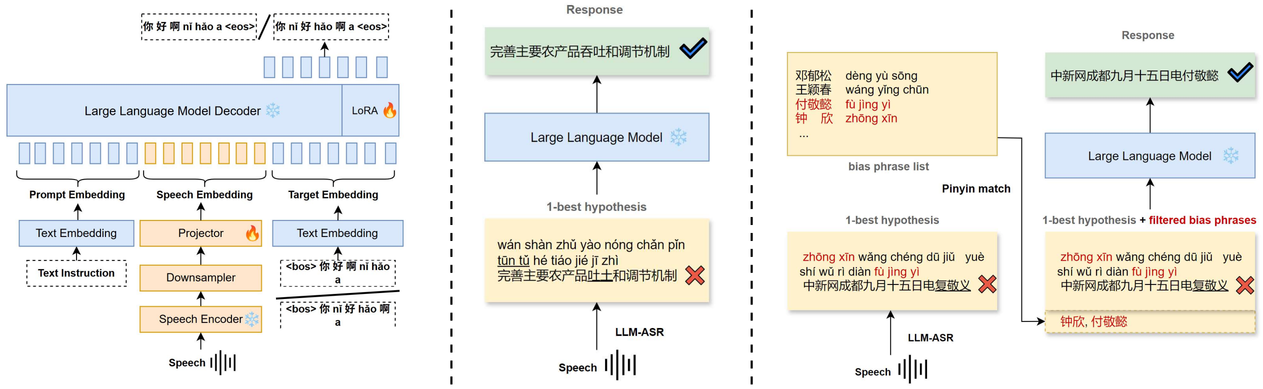

(a) *LLM-ASR model* (b) *PYGEC* (c) *PYGEC-CB*

Figure 1: *The overview of our PYG-ASR framework.*

they all adhere to the same foundational rules. In this paper, we utilize the Pinyin system as presented in *Pypinyin*[1] with the standard tone style, the phonetic tone is on the first letter of the final.

2.2. Model Architecture

As shown in Figure 1 (a), the LLM-ASR model in our approach consists of four main components:

Speech encoder: As shown in Equation 1, for each sample, given speech $\mathbf{X}$, the speech encoder converts each segment of the raw audio signal into a high-dimensional representation $\mathbf{H} = (h_1, h_2, ..., h_T)$.

$$\mathbf{H} = \text{Encoder}(\mathbf{X}) \tag{1}$$

Downsampler: Since speech features usually have much longer lengths than text features, we downsample the speech features $\mathbf{H}$ by concatenating consecutive k frames in the feature dimension, leading to $\mathbf{Z} = (z_1, z_2, ..., z_N)$, where $N = T//k$, this process can be described as shown in Equation 2.

$$z_i = h_{k*i} \oplus h_{k*i+1} \oplus ... \oplus h_{k*i+k-1} \tag{2}$$

projector: The projector is applied to connect the representation generated by the speech encoder with the prompt embedding $\mathbf{E_P}$ of the LLM. It transforms the downsampled speech features $\mathbf{Z}$ into $\mathbf{E_S}$, mapping the output of the speech encoder to the text embedding space of the LLM as shown in Equation 3.

$$\mathbf{E_S} = \text{Linear}(\text{ReLU}(\text{Linear}(\mathbf{Z}))) \tag{3}$$

LLM: Finally, given the speech embedding $\mathbf{E_S}$ and prompt embedding E_P, the generation of the corresponding Pinyin and Chinese character sequence $\mathbf{Y} = (y_1, ..., y_M)$ with the LLM can be formulated as Equation 4.

$$p\left(\mathbf{Y} \mid \mathbf{T}, \mathbf{E_S}; \Theta_{\text{LLM}}\right) = \prod_{m=1}^{M} p\left(y_m \mid \mathbf{Y}_{<m}, \mathbf{E_P}, \mathbf{E_S}; \Theta_{\text{LLM}}\right) \tag{4}$$

[1]https://pypi.org/project/pypinyin/

2.3. Pinyin-Guided Training Process

Although LLMs have been trained on vast amounts of text data and possess strong semantic understanding capabilities, they still lack the ability to predict corresponding text based on pronunciation. To achieve this, we implemented two distinct output strategies, which we denote as PYG-ASR1 and PYG-ASR2 for convenience in later discussions.

PYG-ASR1: Sequential Output of Pinyin and Chinese characters: In the first method, the model generates a complete sequence of Pinyin for the entire speech input first. Then, the Pinyin sequence serves as an intermediate representation to facilitate the subsequent generation of the complete Chinese character sequence. For example, the output would be like this: ”Pinyin sequence: nǐ chī fàn le ma Text: 你吃饭了吗”. This structured approach ensures a coherent transition between the phonetic and written forms, allowing the model to leverage the contextual relationships between Pinyin and Chinese characters effectively. Specifically, we utilize prompt ID 1 as shown in Table 1 for PYG-ASR1 training.

PYG-ASR2: Iterative Output of Pinyin and Chinese characters: The second output method offers a more granular approach by predicting each Chinese character's Pinyin individually. In this method, after generating the Pinyin for a specific character, the model immediately follows this with the output of the corresponding Chinese character. To clarify the output, we use blanks to separate the Pinyin and corresponding Chinese characters. For example, the output would be like this: ”nǐ 你 chī 吃 fàn 饭 le 了 ma 吗”. This iterative approach helps clarify the relationship between the pronunciation and its written form in a character-by-character manner. Specifically, we utilize prompt ID 2 as shown in Table 1 for PYG-ASR1 training.

2.4. Pinyin-Guided Error Correction

In the context of our LLM-ASR model, after generating the Pinyin and corresponding Chinese characters, we employ an error correction phase utilizing a text LLM to further enhance the accuracy of output text. Despite the guidance of Pinyin, cases still persist where the pronunciation is accurately recognized, yet the corresponding Chinese characters are incorrectly generated. For example, an LLM-ASR output might generate ”jiǎo huái 脚槽” while the correct expression should be ”jiǎo huái 脚踝”. The characters ”槽” (cáo) and ”踝” (huái) are pronounced

Table 1: *Prompt templates for PYG-ASR, where "{}" represents a place-holder for the corresponding content*

ID	Prompt
1	*"You are a helpful assistant. Please transcribe the following audio into Pinyin sequence and text in the format of Pinyin: sequence Text: text transcription. Audio: {Audio}"*
2	*"You are a helpful assistant. Please transcribe the following audio into transcription in format of text with pinyin like this: Text with pinyin: 今 jīn 晚 wǎn. Audio: {Audio}"*
3	*"You are a helpful speech chat assistant that can correct speech recognition errors in Chinese characters according to the given Pinyin sequence.\nLet's think step by step:\n1. Check whether the Chinese characters completely correspond to the Pinyin sequence. If not, correct the Chinese characters according to the Pinyin.\n2. Correct the homophone errors in Chinese characters according to the Pinyin sequence.\n3. Do not output any content except for the corrected Chinese characters.\nPinyin sequence: {Pinyin_sequence}, Chinese characters: {Chinese_characters}\nResponse: "*
4	*"You are a helpful speech chat assistant that can correct speech recognition errors in Chinese characters according to the given Pinyin sequence.\nLet's think step by step:\n1. Check whether the Chinese characters completely correspond to the Pinyin sequence. If not, correct the Chinese characters according to the Pinyin.\n2. Correct the homophone errors in Chinese characters according to the Pinyin sequence.\n3. Correct the hotwords errors in Chinese characters according to the Pinyin sequence and part of speech with the given hotwords list.\n4. Do not output any content except for the corrected Chinese characters.\nPinyin sequence: {Pinyin_sequence}, Chinese characters: {Chinese_characters}, hotwords: {hotwords}\nResponse: "*

differently, illustrating a critical error in the hypothesis. Moreover, instances of homophone errors complicate the situation. To address these issues, as shown in Figure 1 (b), PYGEC uses the one-best hypothesis, comprising both the generated Pinyin and corresponding Chinese characters, as input for the text LLM. The process follows a chain-of-thought prompting approach, specifically utilizing prompt ID 3 as shown in Table 1. After processing the input, the LLM, informed both by Pinyin, returns the corrected Chinese characters.

Furthermore, we introduce a contextual biasing version of PYGEC (PYGEC-CB). First, we compare the generated Pinyin with a pre-defined list of bias phrases including contextually relevant terms. The Pinyin matching process identifies any phrases in the list that align with the generated Pinyin, thereby filtering the bias phrases for relevant content. Then, utilizing prompt ID 4 from Table 1, we construct a prompt that includes the filtered bias phrases. After processing the input, the LLM returns the corrected Chinese characters, informed both by the Pinyin and the contextual biasing from the filtered phrases.

3. Experiment Setups

3.1. Datasets and Metrics

We evaluate our method using AISHELL-1, a widely used Mandarin ASR benchmark that contains 170 hours of Mandarin speech. The dataset follows standard splits: 150 hours for training, 10 hours for validation, and 5 hours for testing. We assess ASR performance through CER and Pinyin error rate. To quantify contextual biasing effectiveness, we report B-CER, which measures the error rate for characters in the bias phrases list. Additionally, we use F1-score, precision, and recall to evaluate bias phrase recognition performance.

3.2. Implementation Details

To rigorously assess our method, we select several baseline models, including non-LLM and LLM approaches. The non-LLM baseline is based on the Conformer-based [20] encoder-decoder model, which consists of 12 conformer layers in the encoder and 6 transformer layers in the decoder, both with 512-dimensional inputs and 8 self-attention heads. This model was trained from scratch on ESPNet [21] for 100 epochs.

For our LLM-ASR experiments, we employed Qwen2-7B [1] as our LLM and HuBERT [10] as the speech encoder, which has 315.8 million parameters. We utilize SLAM-ASR [11] as our LLM-ASR baseline, where only the linear projector is trained for the ASR task. Additionally, we explore an enhanced version of SLAM-ASR, which incorporates Low-Rank Adaptation (LoRA) [22] for finetuning LLM. This variant, SLAM-ASR + LoRA, achieves improved alignment and performance [23]. We used the SLAM-LLM toolkit[2] to implement these LLM-ASR experiments. In developing our system, we aligned all settings with those of SLAM-ASR + LoRA, except we modified the LLM's output target. Rather than merely generating the corresponding Chinese characters from the speech input, our system enhances the output by simultaneously generating both Pinyin and Chinese characters. During the processing of speech inputs, HuBERT generates 1024 hidden representations, operating at an output rate of 50Hz. The downsampler stacks consecutive five frames, resulting in an output rate of 10 Hz. The projector is structured as two linear layers, with the hidden layer dimension set to 2048 and then converted to 3048 to align with Qwen's input dimension. For LoRA, we set the rank to 16 and applied it to both the attention and feedforward layers. In the error correction phase, we chose DeepSeek-v3 [2] as the text LLM. All LLM-ASR experiments were conducted on eight NVIDIA A40 GPUs, employing a global batch size of 56 and training the LLM-ASR models for 4 epochs. We use the AdamW optimizer with a max learning rate of 1e-4 without a weight decay. We conduct warmup at the first 1000 steps and then keep the maximum learning rate for training all the time.

4. Results

4.1. Main Results

Table 2 presents the performance of speech recognition on the AISHELL-1 dataset. For the non-LLM baseline, Conformer-AED gets CER of 5.3%/5.9% on the AISHELL-1 dev/test sets. For the LLM baselines, SLAM-ASR gets CER of 4.3%/4.5%. Remarkably, when using LoRA approach to optimize a small number of LLM parameters, the CER reduces to 3.6%/4.0%. These models serve as strong baselines for comparing the performance of PYG-ASR. Our systems demonstrate substantial

[2]https://github.com/X-LANCE/SLAM-LLM

improvements over the baselines. The sequential output configuration consistently outperformed the iterative output when PYGEC was not applied. When PYGEC was integrated, we observed a notable enhancement in ASR accuracy, resulting in CER of 2.8/3.0 for both configurations. Moreover, we considered the Pinyin error rate for evaluating the phonetic accuracy, helping to further assess the system's effectiveness. The Pinyin error rates of the PYG-ASR models displayed significant reductions as well, which correlate with the improvements seen in CER. The PYG-ASR1 model achieved a Pinyin error rate of 1.6%/1.9%, and PYG-ASR2 gets 1.7%1.9%. In comparison, the best baseline model, SLAM-ASR + LoRA, reported CER of 2.5%/2.9% and Pinyin error rate of 2.5%/2.9%. In summary, our proposed PYG-ASR models not only enhance character recognition but also improve pronunciation accuracy.

Table 2: *CER(%)/Pinyin error rate(%) of various models on AISHELL-1 Dev and Test.*

Model	Trainable Params(M)	AISHELL-1 Dev	AISHELL-1 Test
Baselines			
Conformer-AED	114.97	5.3/3.2	5.9/3.6
SLAM-ASR	17.83	4.3/3.1	4.5/3.4
SLAM-ASR + LoRA	20.35	3.6/2.5	4.0/2.9
Ours			
PYG-ASR1	20.35	3.3/**1.6**	3.8/**1.9**
+ PYGEC	20.35	**2.8**/1.9	**3.0**/2.1
PYG-ASR2	20.35	4.1/1.7	4.3/**1.9**
+ PYGEC	20.35	**2.8**/1.7	**3.0**/2.0

4.2. Contextual Biasing

Table 3 presents a performance comparison among various contextual ASR models using an artificial lists on AISHELL-1 test set, with the bias phrase list size set to 186. This comparison highlights the significant impact of PYGEC-CB. PYGEC-CB achieved a B-CER of 12.55% and an F1-score of 88.62%. Compared to results reported in [24], our method attains even better performance without finetuning or additional specialized designs. This demonstrates that contextual biasing can be effectively achieved during the error correction phase with Pinyin guidance.

Table 3: *Results for contextual biasing evaluation in the following format: B-WER(F1-score/Recall/Precision), 'cb' means contextual biasing.*

Method	B-CER(F1/Recall/Precision)
CPPN [25]	38.90 (47.40/ 31.40/ 96.80)
+ shallow fusion	20.80 (78.40/ 66.20/ 96.00)
CBCB [24]	20.30 (79.50/ 66.20/ 99.50)
+ shallow fusion	15.50 (86.20/ 77.00/ 97.80)
PYG-ASR1	24.70 (71.26/ 55.61/ 99.20)
+ PYGEC	23.26 (73.55/ 58.43/ 99.24)
+ PYGEC-CB	**12.55** (88.62/ 82.77/ 95.36)

4.3. Case Study

Table 4 presents two cases to illustrate the performance of our method.

In Case 1, the baseline models exhibited varying levels of accuracy. In contrast, our PYG-ASR model produced correct Pinyin but generated inconsistent Chinese characters initially. However, after applying PYGEC, the performance improved significantly, resulting in a best CER of 5%. This demonstrates the effectiveness of integrating the error correction phase.

In case 2, all models struggled to accurately transcribe the name "付敬懿" without the aid of contextual biasing. To address this, we performed Pinyin matching and obtained the filtered phrases "付敬懿" and "钟欣". These phrases were used to prompt the text LLM. After the application of PYGEC-CB, "复敬义" was successfully corrected to "付敬懿".

Table 4: *Case analysis of different models. Baseline 1/2/3 are Conformer-AED, SLAM-ASR, and SLAM-ASR + LoRA, respectively.*

Method	Uttrance	CER%
	case1	
GT	拍照踩死植物水杉栈道仙境拉铁丝网	-
baseline1	拍照才死植物水山站到先进拉铁丝网	30.0
baseline1	拍照踩死植物水杉栈道探险拉铁丝网	10.0
baseline3	拍照踩死职务水杉栈道严禁拉铁丝网	20.0
PYG-ASR1	拍照踩死职务水杉站道先锋拉铁丝网	25.0
+ PYGEC	拍照踩死植物水杉站道仙境拉铁丝网	5.0
PYG-ASR2	拍照采死植物水山战道先进拉铁丝网	25.0
+ PYGEC	拍照踩死植物水杉栈道先进拉铁丝网	10.0
	case2	
GT	中新网成都九月十五日电付敬懿十五日	-
baseline1	中新网成都九月十五日电付敬意十五日	5.9
baseline2	中新网成都九月十五日电附敬一丹十五日	23.5
baseline3	中新网成都九月十五日电复靖逸十五日	17.6
PYG-ASR1	中新网成都九月十五日电复敬义十五日	11.8
+ PYGEC-CB	中新网成都九月十五日电付敬懿十五日	0.0
PYG-ASR2	中新网成都九月十五日电复敬义十五日	11.8
+ PYGEC-CB	中新网成都九月十五日电付敬懿十五日	0.0

5. Conclusion

In this paper, we proposed a Pinyin-guided Chinese ASR framework to enhance the LLM-ASR performance. Our method explored introducing Pinyin prediction into Chinese LLM-ASR models. We explicitly modeled the pronunciation features to enhance the model's ability to generate transcripts based on pronunciation features by mapping acoustic features to Pinyin tokens and Chinese character tokens simultaneously. Furthermore, we also proposed a simple yet effective error correction technique with a chain of thought to further enhance the robustness and performance.

Additionally, we effectively completed contextual bias in the error correction phrase. For future works, we intend to conduct larger-scale experiments and explore a more profound integration of Pinyin information into LLM-ASR models.

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